# Exercise Sheet 1

All tangent spaces are real. A tangent vector is represented by a smooth curve through the stated point. Use the matrix commutator for tangent brackets. Exercise 1.6 is an optional extension; no other exercise depends on it.

**Exercise 1.1 (The circle).** Regard $\mathrm{U}(1)$ as the unit circle in $\mathbb{C}$.

(i) Differentiate $\overline{z(t)}z(t) = 1$ at $z(0) = 1$.

(ii) For any $a \in \mathbb{R}$, construct a unit-circle curve with $z(0) = 1$ and $z'(0) = ia$. Find the tangent bracket.

**Exercise 1.2 (Tangent vectors at a rotation).** Let $g \in \mathrm{SO}(n)$. Prove that

$$T_g \,\mathrm{SO}(n) = \{gX : X^{\mathsf{T}} = -X\}.$$

Give a basis of $\mathfrak{so}(3)$ and its dimension. The matrix unit $E_{ij}$ has one in position $(i, j)$ and zero elsewhere.

**Exercise 1.3 (A bracket in so(3)).** Set

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \qquad J_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Compute $[J_1, J_2]$. Use the cyclic pattern to compare $\mathfrak{so}(3)$ with the basis $(A, B, C)$ of $\mathfrak{su}(2)$ in the lecture. Find a real Lie algebra isomorphism by specifying its values on $A, B, C$.

**Exercise 1.4 (The real dimension of U(n)).** Apply the regular level-set theorem to

$$F\colon M_{n\times n}(\mathbb{C}) \longrightarrow \{S : S^* = S\}, \qquad F(U) = U^*U.$$

Both spaces are real vector spaces. Prove that $dF_U$ is onto at a unitary matrix. Find $\dim_{\mathbb{R}} \mathrm{U}(n)$ and its tangent algebra at $I$.

**Exercise 1.5 (Complexification).** Let $Z \in \mathfrak{sl}_2(\mathbb{C})$. Show that it has a unique decomposition

$$Z = X + iY, \qquad X, Y \in \mathfrak{su}(2).$$

Deduce directly that the complex-linear map $\iota\colon \mathfrak{su}(2)_{\mathbb{C}} \to \mathfrak{sl}_2(\mathbb{C})$ of Lecture 1, given by $\iota(X \otimes a) = aX$, is an isomorphism.

**Exercise 1.6 (A group not determined by its tangent algebra).** Give the two-element group $C_2 = \{1, s\}$ its discrete topology. Compare the Lie groups $\mathbb{R}$ and $\mathbb{R} \times C_2$, with the additive law in the first factor. Find their tangent algebras and prove that they are not isomorphic as Lie groups. You may use connectedness of $\mathbb{R}$.